



JAKARTA EE CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & APP SERVER

Installation

```
WildFly / GlassFish / Payara / OpenLiberty
<packaging>war</packaging>
<dep>jakarta.jakartaee-api 10</dep> (provided)
@ApplicationPath("/api") class AppConfig exte...
Deploys as .war on servlet container or app server
```

04. SERVLETS

Workflow

```
@WebServlet("/hello")
public class HelloServlet extends HttpServlet{
protected void doGet(HttpServletRequest req,
HttpServletResponse res){
res.getWriter().write("Hello");
}
}
```

02. CDI BEANS

Architecture

```
@RequestScoped / @SessionScoped / @Applicatio...
@Dependent (default lifecycle follows owner)
@Inject field, constructor, or setter
@Named("myBean") EL accessible
@Produces factory method for CDI
@Qualifier disambiguate multiple impls
```

05. JPA INTEGRATION

CRUD API

```
@PersistenceContext EntityManager em;
@Stateless // or @RequestScoped + @Transacti...
public List<User> findAll(){
return em.createQuery("FROM User",User.class)...
}
```

JTA transactions managed by container

03. JAX-RS ENDPOINTS

YAML / Config

```
@Path("/users") @ApplicationScoped
public class UserResource {
@GET @Path("/{id}") @Produces(APPLICATION_JS0...
public User getUser(@PathParam("id") Long id)...
@POST @Consumes(APPLICATION_JSON)
public Response create(User u){...}
}
```

06. BEAN VALIDATION

Integration

```
@NotNull @Size(min=2,max=50) String name;
@email String email;
@Min(0) int age;
@Valid in JAX-RS params triggers validation
Validator v = Validation.buildDefaultValidato...
Set<ConstraintViolation<T>> errs = v.validate...
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. JMS MESSAGING

Hardening

```
@Inject @JMSConnectionFactory("jms/cf")
JMSContext ctx;
@Resource(lookup="jms/queue") Queue q;
ctx.createProducer().send(q, "message");
@MessageDriven(activationConfig={...})
void onMessage(Message m){...} // MDB
```

10. WEBSOCKET

Unit Tests

```
@ServerEndpoint("/ws/chat")
public class ChatEndpoint{
    @OnOpen void onOpen(Session s){}
    @OnMessage void onMsg(String msg, Session s){}
    @OnClose void onClose(Session s){}
}
```

Full-duplex communication over HTTP upgrade

08. EJB BASICS

Observability

```
@Stateless    no state, pooled
@Stateful     conversational state per client
@Singleton    @Startup    one instance
```

Container manages transactions on @Stateless by default

```
@TransactionAttribute(REQUIRED/REQUIRES_NEW/N...
```

EJBs deprecated in favor of CDI for new projects

11. SECURITY

Commands

```
@DeclareRoles({"admin","user"})
@RolesAllowed("admin") on method or class
@PermitAll / @DenyAll
SecurityContext.isUserInRole("admin")
@WebSecurityDef (Soteria 2.0 Jakarta Securit...
@loginConfig(authMethod="FORM") in web.xml
```

09. INTERCEPTORS

Optimization

```
@Interceptor @LoggingBinding
public class LoggingInterceptor{
    @AroundInvoke
    public Object log(InvocationContext ctx) thro...
    log.info("Calling "+ctx.getMethod().getName()...
    return ctx.proceed();
}
```

12. COMMON GOTCHAS

Warning

EJB must not be new'd directly use @Inject
CDI @RequestScoped doesn't work in async/scheduler
JAX-RS providers need @Provider on the class
persistence.xml required even with Spring-style JPA
Application server version must match API version